Glossary

212 terms from every unit. Search, or jump by letter. Each entry links back to the unit where it is taught.

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Unit 1: Your First Python Program

Assignment

Storing a value under a name using `=`. The right side is computed first.

Unit 1

Comment

Text after `#` that Python ignores. Used to explain intent to human readers.

Unit 1

Interpreter

The program that reads Python code and runs it. Python code runs line by line through the interpreter.

Unit 1

Keyword

A word reserved by the language (for example `if`, `for`, `True`). It cannot be used as a variable name.

Unit 1

NameError

The error raised when code uses a name that has not been assigned.

Unit 1

print()

A built-in function that writes its arguments to the output, separated by spaces, followed by a newline.

Unit 1

snake_case

Naming style with lowercase words joined by underscores, such as `total_price`. Python's convention for variables and functions.

Unit 1

Statement

One complete instruction the interpreter runs, such as `print("hi")` or `x = 3`.

Unit 1

SyntaxError

The error raised when code breaks the grammar of the language, such as a missing quotation mark or parenthesis.

Unit 1

Variable

A name that refers to a value stored in memory. The value can be changed by assigning a new one.

Unit 1

Unit 2: Data Types and Strings

bool

Boolean: the value `True` or `False`. Named for George Boole.

Unit 2

Casting

Converting a value from one type to another, for example `int("42")`.

Unit 2

Data type

A category of value (int, float, str, bool) that determines what operations are valid on it.

Unit 2

f-string

A string literal beginning with `f` that embeds expressions in braces, such as `f"{x}"`.

Unit 2

float

Floating-point number: a number with a decimal point. Stored approximately.

Unit 2

Immutable

A value that cannot be changed after it is created. Strings, integers, and tuples are immutable.

Unit 2

Index

The position of an item in a sequence. Counting starts at 0.

Unit 2

int

Integer: a whole number with no decimal point.

Unit 2

Method

A function that belongs to an object and is called with a dot, such as `name.upper()`.

Unit 2

Slice

A part of a sequence written `seq[start:stop]`. The stop position is excluded.

Unit 2

str

String: an immutable sequence of characters written in quotes.

Unit 2

type()

A built-in function that returns the type of a value, such as `<class 'int'>`.

Unit 2

Unit 3: Boolean Logic and Truth Tables

Boolean

A value that is either `True` or `False`.

Unit 3

Chained comparison

A comparison such as `1 < x < 10`, which Python evaluates as two comparisons joined by `and`.

Unit 3

Comparison operator

An operator that compares two values and returns a boolean: `==`, `!=`, `<`, `>`, `<=`, `>=`.

Unit 3

Logical operator

`and`, `or`, and `not`, which combine or invert boolean values.

Unit 3

Operator precedence

The fixed order in which operators are evaluated. Parentheses override it.

Unit 3

Short-circuit evaluation

Stopping a boolean expression as soon as its result is known, so later parts are never evaluated.

Unit 3

Truth table

A table listing every combination of inputs and the resulting output of a logic expression.

Unit 3

Truthy / falsy

Values that count as True or False in a condition. Falsy values include `0`, `""`, `[]`, and `None`.

Unit 3

XOR (exclusive or)

True when exactly one of two inputs is True. In Python, `a != b` on booleans.

Unit 3

Unit 4: Conditionals: Making Decisions

Block

A group of statements that run together, introduced by a colon and indented beneath a statement.

Unit 4

Branch

One path through a conditional. Each `if`, `elif`, and `else` defines a branch.

Unit 4

Condition

An expression that evaluates to True or False and decides whether a block runs.

Unit 4

Conditional expression

A one-line choice written `a if condition else b`.

Unit 4

elif

Short for "else if". Tests another condition only if the previous ones were False.

Unit 4

Indentation

Leading spaces that show which lines belong to a block. Python requires consistent indentation.

Unit 4

Nested conditional

A conditional placed inside another conditional's block.

Unit 4

Unit 5: Loops: Repeating Work

Accumulator

A variable that collects a running result, such as a total, across loop iterations.

Unit 5

break

A statement that exits the nearest loop immediately.

Unit 5

continue

A statement that skips the rest of the current iteration and moves to the next.

Unit 5

for loop

A loop that assigns each item of a sequence to a variable in turn.

Unit 5

Infinite loop

A loop whose condition never becomes False, so it never ends unless interrupted.

Unit 5

Iteration

One pass through the body of a loop.

Unit 5

Loop

A statement that repeats a block of code.

Unit 5

Off-by-one error

A bug caused by a range or index that is one too high or too low.

Unit 5

range()

A built-in that produces a sequence of integers from `start` up to but not including `stop`.

Unit 5

while loop

A loop that repeats while its condition is True. The condition is checked before each pass.

Unit 5

Unit 6: Functions

Argument

The actual value passed to a function when it is called.

Unit 6

Default argument

A parameter with a preset value, such as `rate=0.07`, used when the caller does not supply one.

Unit 6

Docstring

A string literal placed as the first line of a function body to document what it does.

Unit 6

Function

A named, reusable block of code, defined with `def` and run by calling it.

Unit 6

Global variable

A variable defined at the top level of a program, visible to the code that follows it.

Unit 6

Keyword argument

An argument passed by name, such as `greet(name="Sam")`.

Unit 6

Local variable

A variable that exists only inside the function where it was created.

Unit 6

None

Python's special value meaning "no value". Functions without a `return` produce it.

Unit 6

Parameter

A variable name in a function definition that receives a value when the function is called.

Unit 6

return

A statement that ends a function and sends a value back to the caller.

Unit 6

Unit 7: Lists

Aliasing

When two names refer to the same object, so a change through one name is visible through the other.

Unit 7

enumerate()

A built-in that gives each item of a sequence together with its index, for use in a loop.

Unit 7

Index

The position of an item in a list, starting at 0. Negative indexes count from the end.

Unit 7

IndexError

The error raised when an index is outside the range of a list.

Unit 7

List

An ordered, mutable collection of values written in square brackets.

Unit 7

List comprehension

A compact expression that builds a list from a loop, such as `[n * n for n in range(5)]`.

Unit 7

Method

A function attached to an object. `append`, `insert`, `pop`, and `sort` are list methods.

Unit 7

Mutable

A value that can be changed after it is created. Lists and dictionaries are mutable.

Unit 7

Slice

A new list made from a range of positions, written `items[start:stop]`.

Unit 7

Unit 8: Tuples and Sets

Difference

The items in the first set that are not in the second. Written `a - b`.

Unit 8

Immutable

Cannot be changed after creation. Tuples, strings, and numbers are immutable.

Unit 8

Intersection

The set of items in both sets. Written `a & b`.

Unit 8

Membership test

Checking whether a value is in a collection with the `in` operator.

Unit 8

Set

An unordered collection of unique values, written in curly braces or created with `set()`.

Unit 8

Tuple

An immutable, ordered collection written in parentheses, such as `(4, 7)`.

Unit 8

Union

The set of items in either of two sets. Written `a | b`.

Unit 8

Unpacking

Assigning each item of a sequence to a separate name in one statement, such as `x, y = point`.

Unit 8

Unit 9: Dictionaries

.get()

A dictionary method that returns the value for a key, or a default if the key is missing.

Unit 9

.items()

A dictionary method that yields key-value pairs, usually unpacked in a loop as `for k, v in d.items()`.

Unit 9

Counter pattern

Using a dictionary to count occurrences: `counts[item] = counts.get(item, 0) + 1`.

Unit 9

Dictionary

A collection that maps unique keys to values, written in curly braces with `key: value` pairs.

Unit 9

Key

The label used to look up a value in a dictionary. Keys must be immutable and unique.

Unit 9

KeyError

The error raised when reading a key that is not in the dictionary.

Unit 9

Nested dictionary

A dictionary that contains another dictionary or list as a value.

Unit 9

Value

The data stored under a key. Values can be any type.

Unit 9

Unit 10: Libraries and Modules

__name__

A built-in variable. It equals `"__main__"` when a file is run directly, and the module name when imported.

Unit 10

Alias

A shorter name given to a module on import, such as `import numpy as np`.

Unit 10

import

A statement that loads a module so its names can be used.

Unit 10

Library

A collection of modules that provides a set of related tools, such as the standard library, NumPy, or pandas.

Unit 10

Module

A Python file (`.py`) whose functions and variables can be imported into another program.

Unit 10

pip

The package installer for Python. It downloads third-party libraries from the Python Package Index.

Unit 10

Seed

A starting value for a random number generator. The same seed produces the same sequence every run.

Unit 10

Standard library

The modules that come with every Python installation, such as `math`, `random`, and `json`.

Unit 10

Unit 11: Errors and Exceptions

else (try)

A block that runs only when the try block completed without an exception.

Unit 11

Exception

An error signal raised when something goes wrong during execution. Unhandled exceptions stop the program.

Unit 11

finally

A block that always runs after try/except, used for cleanup.

Unit 11

raise

A statement that raises an exception on purpose.

Unit 11

Traceback

The report Python prints for an unhandled exception, listing the call path and ending with the exception type and message.

Unit 11

try / except

A block that runs code and jumps to a handler if a named exception is raised.

Unit 11

TypeError

Raised when an operation is applied to a value of the wrong type.

Unit 11

ValueError

Raised when an argument has the right type but an inappropriate value.

Unit 11

Unit 12: Advanced Functions and Comprehensions

**kwargs

A parameter that collects extra keyword arguments into a dictionary.

Unit 12

*args

A parameter that collects extra positional arguments into a tuple.

Unit 12

Base case

The condition that stops a recursive function from calling itself again.

Unit 12

Comprehension

A compact expression that builds a list, dictionary, or set from a loop and an optional condition.

Unit 12

filter

A built-in that keeps the items for which a function returns True.

Unit 12

Generator expression

A comprehension in parentheses that produces values lazily, one at a time.

Unit 12

Iterator

An object that produces items one at a time and can be looped over once.

Unit 12

lambda

A one-expression, unnamed function, written `lambda x: x * 2`.

Unit 12

map

A built-in that applies a function to each item of a sequence, returning an iterator.

Unit 12

Recursion

A function that calls itself on a smaller problem until it reaches a base case.

Unit 12

zip

A built-in that pairs up items from several sequences, stopping at the shortest.

Unit 12

Unit 13: Files, CSV, and JSON

CSV

Comma-separated values: plain text with one record per line and commas between fields.

Unit 13

Encoding

The rule for turning characters into bytes, such as UTF-8. Mismatched encodings garble text.

Unit 13

File object

The object returned by `open()` that provides read and write methods for a file.

Unit 13

JSON

JavaScript Object Notation: a text format for nested data made of objects, arrays, strings, numbers, booleans, and null.

Unit 13

Mode

The letter(s) passed to `open()` that set reading, writing, or appending, such as `"r"`, `"w"`, or `"a"`.

Unit 13

Relative path

A file path measured from the folder the program runs in, rather than from the root of the disk.

Unit 13

Serialize

Convert data in memory into a format that can be stored or sent, such as a JSON string.

Unit 13

with statement

A block that acquires a resource and guarantees it is released when the block ends.

Unit 13

Unit 14: Classes and Objects

__init__

The special method that sets up a new object when it is created.

Unit 14

__str__

The special method that returns the text shown by `print()` and `str()`.

Unit 14

Attribute

A variable that belongs to an object, read with a dot, such as `maya.name`.

Unit 14

Class

A blueprint that defines the attributes and methods shared by a kind of object.

Unit 14

Dataclass

A class decorated with `@dataclass` that generates the constructor and printable form from its field annotations.

Unit 14

Decorator

A line starting with `@` placed above a function or class that wraps or modifies it.

Unit 14

Inheritance

A class that takes over the attributes and methods of a parent class, written `class Child(Parent):`.

Unit 14

Method

A function defined inside a class. It receives the object as `self`.

Unit 14

Object / instance

A specific value created from a class. Each instance has its own attributes.

Unit 14

Polymorphism

Using one interface on different kinds of objects, where each responds in its own way.

Unit 14

self

The conventional name of the first parameter of a method, referring to the object the method was called on.

Unit 14

super()

A way to call a method from the parent class, such as the parent's `__init__`.

Unit 14

Unit 15: The Standard Library Toolkit

Counter

A dictionary subclass from `collections` that counts how often each item appears.

Unit 15

defaultdict

A dictionary that creates a default value (such as an empty list) for any missing key.

Unit 15

itertools

A standard-library module of fast, memory-efficient tools for looping and combining items.

Unit 15

Path

An object from `pathlib` that represents a file or folder path and offers methods to read, write, and inspect it.

Unit 15

Raw string

A string literal prefixed with `r` that treats backslashes literally, commonly used for regex patterns.

Unit 15

Regular expression

A pattern that describes a set of strings, used to search, match, and replace text.

Unit 15

strftime / strptime

Methods that format a date as text (`strftime`) or parse text into a date (`strptime`).

Unit 15

timedelta

A duration: the difference between two dates or times.

Unit 15

Unit 16: NumPy: Arrays and Fast Math

Axis

The dimension an aggregation runs along. `axis=0` runs down rows; `axis=1` runs across columns.

Unit 16

Boolean mask

An array of True and False values, used to select elements that meet a condition.

Unit 16

Broadcasting

The rules NumPy uses to combine arrays of different shapes by stretching the smaller one.

Unit 16

dtype

The data type of every element in an array, such as `int64` or `float64`.

Unit 16

Matrix product

The `@` operator, which multiplies two arrays using rows of the first and columns of the second.

Unit 16

ndarray

NumPy's n-dimensional array: a block of same-typed values with a shape.

Unit 16

Reshape

Changing an array's shape while keeping the same data and the same number of elements.

Unit 16

shape

A tuple giving the size of each dimension, such as `(2, 3)` for two rows and three columns.

Unit 16

Vectorization

Writing an operation on a whole array at once instead of looping over its elements.

Unit 16

Unit 17: pandas: Tables You Can Question

DataFrame

A two-dimensional labeled table in pandas, with named columns that can each have a different type.

Unit 17

groupby

A method that splits rows into groups by the values of one or more columns, so you can summarize each group.

Unit 17

Index

The labels for the rows of a DataFrame or Series. The default is 0, 1, 2, and so on.

Unit 17

loc / iloc

`loc` selects by label; `iloc` selects by integer position.

Unit 17

NaN

Not a Number: pandas' marker for a missing value in a numeric column.

Unit 17

read_csv

The pandas function that loads a CSV file (or text) into a DataFrame.

Unit 17

Series

A one-dimensional labeled array, such as a single column of a DataFrame.

Unit 17

value_counts

A method that counts how often each distinct value appears in a column.

Unit 17

Unit 18: Visualizing Data with matplotlib

Axes

A single plotting area inside a figure, with its own x and y axes, title, and data.

Unit 18

Bin

A range of values grouped together in a histogram.

Unit 18

Figure

The whole picture that matplotlib draws, which can hold one or more axes.

Unit 18

Histogram

A chart that groups numeric values into bins and shows how many fall in each bin.

Unit 18

matplotlib

The standard Python library for drawing charts. Its `pyplot` module is usually imported as `plt`.

Unit 18

Scatter plot

A chart of points, one per observation, that shows the relationship between two numbers.

Unit 18

subplots

A function that creates a grid of axes in one figure.

Unit 18

Truncated axis

An axis that does not start at zero, which can exaggerate differences between bars.

Unit 18

Unit 19: Machine Learning with scikit-learn

Accuracy

The fraction of predictions that match the true labels.

Unit 19

Confusion matrix

A table of true labels against predicted labels, showing which mistakes the model makes.

Unit 19

Data leakage

When information from the test set influences training, making the measured score too optimistic.

Unit 19

Feature

An input variable used to make a prediction, usually a column in the feature table.

Unit 19

Label / target

The value a supervised model is trained to predict.

Unit 19

Overfitting

When a model fits the training data, including its noise, so well that it performs worse on new data.

Unit 19

Pipeline

A scikit-learn object that chains preprocessing steps and a model so they are fitted and applied in order.

Unit 19

R^2

A regression score where 1.0 is a perfect fit and 0 means no better than predicting the average.

Unit 19

Supervised learning

Learning from examples that include the correct label.

Unit 19

Train/test split

Dividing data into a set used for fitting and a held-back set used only for evaluation.

Unit 19

Unsupervised learning

Finding structure in data that has no labels, such as clusters.

Unit 19

Unit 20: PyTorch: Tensors, Gradients, and Neural Networks

Activation function

A non-linear function applied after a layer, such as ReLU, so the network can learn curves.

Unit 20

Autograd

PyTorch's automatic differentiation: it records operations and computes gradients on `backward()`.

Unit 20

Device

Where a tensor lives: `cpu` or an accelerator such as `cuda`.

Unit 20

dtype

The number type stored in a tensor, such as `float32` or `int64`.

Unit 20

Gradient

The rate at which an output changes as a parameter changes. Training moves parameters against the gradient.

Unit 20

Logits

The raw, unnormalized scores a classifier outputs before a softmax or other probability step.

Unit 20

Neuron

A unit that computes a weighted sum of its inputs plus a bias, then applies an activation function.

Unit 20

nn.Module

The base class for PyTorch models. Layers are defined in `__init__` and the data flow in `forward`.

Unit 20

Parameters

The learned numbers of a model, such as weights and biases.

Unit 20

Tensor

A multi-dimensional array in PyTorch that can run on a GPU and track gradients.

Unit 20

Unit 21: Training a PyTorch Model

Batch / mini-batch

A subset of the training data processed together in one step of the loop.

Unit 21

CrossEntropyLoss

The standard loss for classification. It takes raw logits and integer class labels.

Unit 21

Epoch

One full pass of the training data through the model.

Unit 21

Learning rate

The size of each update step taken by the optimizer.

Unit 21

Loss function

A formula that measures how wrong the predictions are. Training minimizes it.

Unit 21

model.eval()

Switches a model to evaluation behavior, turning off training-only behaviors such as dropout.

Unit 21

Optimizer

The algorithm that updates the weights from their gradients, such as SGD or Adam.

Unit 21

state_dict

A dictionary of a model's learned parameters, used to save and load it.

Unit 21

Stratify

Splitting data so each class keeps the same proportion in the training and test sets.

Unit 21

torch.no_grad()

A context manager that turns off gradient tracking for faster, lighter evaluation.

Unit 21

Unit 22: Capstone and Responsible Data Work

Aggregation

Reporting summaries (counts, averages) instead of individual records, to protect privacy.

Unit 22

Baseline

A simple reference model, such as always predicting the most common class, that any real model must beat.

Unit 22

Fairness (group metrics)

Checking that a model's performance is comparable across the groups it affects.

Unit 22

FERPA

The US federal law protecting the privacy of student education records.

Unit 22

Leakage

When a model is trained on information that would not be available at prediction time, inflating its score.

Unit 22

Model card

A short document describing a model's purpose, data, evaluation, limits, and owner.

Unit 22

Unsupervised learning

Learning structure from unlabeled data, such as clustering similar examples together.

Unit 22

Unit 23: Working Like a Python Developer

assert

A statement that raises `AssertionError` when its condition is False, used for quick checks and tests.

Unit 23

Generator

A function that uses `yield` to produce values one at a time, pausing between them.

Unit 23

global / nonlocal

Keywords that let a function assign a variable from the module level (`global`) or an enclosing function (`nonlocal`).

Unit 23

input()

A built-in that shows a prompt and returns the user's typed text as a string.

Unit 23

is

The identity operator. True when two names refer to the same object in memory.

Unit 23

match statement

A statement that compares a value against patterns and runs the first pattern that fits (Python 3.10+).

Unit 23

Notebook

A document of code and text cells, each runnable in a shared session. Jupyter and Google Colab both use notebooks.

Unit 23

pip

The package installer for Python. Installs libraries from the Python Package Index.

Unit 23

unittest

The standard-library testing framework. Tests are methods named `test_...` inside a `TestCase` class.

Unit 23

Virtual environment

An isolated folder of installed packages for one project, created with `python -m venv`.

Unit 23

yield

A statement that returns a value from a generator and pauses until the next value is requested.

Unit 23

A–Z

A

Accumulator · Accuracy · Activation function · Aggregation · Alias · Aliasing · Argument · assert · Assignment · Attribute · Autograd · Axes · Axis

B

Base case · Baseline · Batch / mini-batch · Bin · Block · bool · Boolean · Boolean mask · Branch · break · Broadcasting

C

Casting · Chained comparison · Class · Comment · Comparison operator · Comprehension · Condition · Conditional expression · Confusion matrix · continue · Counter · Counter pattern · CrossEntropyLoss · CSV

D

Data leakage · Data type · Dataclass · DataFrame · Decorator · Default argument · defaultdict · Device · Dictionary · Difference · Docstring · dtype · dtype

E

elif · else (try) · Encoding · enumerate() · Epoch · Exception

F

f-string · Fairness (group metrics) · Feature · FERPA · Figure · File object · filter · finally · float · for loop · Function

G

Generator · Generator expression · global / nonlocal · Global variable · Gradient · groupby

H

Histogram

I

Immutable · Immutable · import · Indentation · Index · Index · Index · IndexError · Infinite loop · Inheritance · input() · int · Interpreter · Intersection · is · Iteration · Iterator · itertools

J

JSON

K

Key · KeyError · Keyword · Keyword argument

L

Label / target · lambda · Leakage · Learning rate · Library · List · List comprehension · loc / iloc · Local variable · Logical operator · Logits · Loop · Loss function

M

map · match statement · matplotlib · Matrix product · Membership test · Method · Method · Method · Mode · Model card · model.eval() · Module · Mutable

N

NameError · NaN · ndarray · Nested conditional · Nested dictionary · Neuron · nn.Module · None · Notebook

O

Object / instance · Off-by-one error · Operator precedence · Optimizer · Overfitting

P

Parameter · Parameters · Path · pip · pip · Pipeline · Polymorphism · print()

R

raise · range() · Raw string · read_csv · Recursion · Regular expression · Relative path · Reshape · return · R^2

S

Scatter plot · Seed · self · Serialize · Series · Set · shape · Short-circuit evaluation · Slice · Slice · snake_case · Standard library · state_dict · Statement · str · Stratify · strftime / strptime · subplots · super() · Supervised learning · SyntaxError

T

Tensor · timedelta · torch.no_grad() · Traceback · Train/test split · Truncated axis · Truth table · Truthy / falsy · try / except · Tuple · type() · TypeError

U

Union · unittest · Unpacking · Unsupervised learning · Unsupervised learning

V

Value · value_counts · ValueError · Variable · Vectorization · Virtual environment

W

while loop · with statement

X

XOR (exclusive or)

Y

yield

Z

zip